CS5760: Cryptanalysis of DES and DES-like Iterated Cryptosystems

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February 3, 2025

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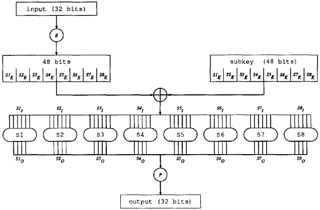
   - *Invariant* in key mixing with subkey S_K to get $S_I = S_E \oplus S_K$.
   - *Linear* in permutation P on S_O after S boxes.
   - *Invariant* in XOR operation connecting rounds.



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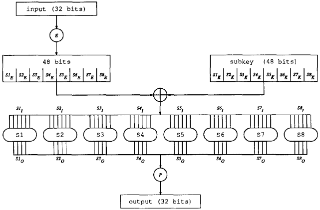
4. S boxes are *nonlinear*. Probability analysis performed between input and output XOR.



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3. This joint PMF can reduce the number of possible (sub)keys. Used to drive choice for the plaintext XOR.
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    - Given Si'_I and Si'_O, we can narrow down Si_K to a few possibilities.

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   - $\approx 80\%$ entries are non-zero/possible for each S box (some have lesser percentages).
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4. i^{th} S box contributes probability p_i for $Si'_I \to Si'_O$.
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   - Over n rounds, $P = \prod_{i=1}^{n} P_i$.

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Desirable for cryptanalysis: high P with large n.

Characteristic

Formalizes notion of high-probability plaintext XORs.

Definition 1 (Characteristic)

An *n-round chracteristic* is a tuple $\Omega = (\Omega_P, \Omega_\Lambda, \Omega_T)$ where $\Omega_P = (L', R')$ and $\Omega_T = (l', r')$ are m bit numbers, $\Omega_\Lambda = (\Lambda_1, \dots, \Lambda_n)$, $\Lambda_i = (\lambda_I^i, \lambda_O^i)$ and $\lambda_I^i, \lambda_O^i, L', R', l', r'$ are $\frac{m}{2}$ bit numbers and m is the block size of the cryptosystem satisfying

$$\lambda_I^1 = R' \tag{1}$$
$$\lambda_I^2 = L' \oplus \lambda_O^1 \tag{2}$$
$$\lambda_I^n = r' \tag{3}$$
$$\lambda_I^{n-1} = l' \oplus \lambda_O^n \tag{4}$$
$$\forall\ 1 < i < n,\ \lambda_O^i = \lambda_I^{i-1} \oplus \lambda_I^{i+1} \tag{5}$$

Characteristic

Definition 2 (Right Pair)

A *right pair with respect to an n-round characteristic* $\Omega = (\Omega_P, \Omega_\Lambda, \Omega_T)$ *and an independent key* K is a pair for which $P' = \Omega_P$ and for each round i of the first n rounds of the encryption of the pair using K the input XOR of the i^{th} round equals λ_I^i and the output XOR of the F function equals λ_O^i. Pairs that do not satisfy these conditions are called *wrong pairs*.

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Definition 3 (Probability of a Round of a Characteristic)

Round i of an n-round characteristic Ω has probability p_i^Ω if $\lambda_I^i \to \lambda_O^i$ with probability p_i^Ω by the F function.

Probability of a Characteristic

Definition 4 (Probability of a Characteristic)

An n-round characteristic Ω has probability p^{Ω} given by

$$p^{\Omega} = \prod_{i=1}^{n} p_i^{\Omega} \tag{6}$$

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Theorem 5 (Probability of a Characteristic and Right Pairs)

The formally defined probability of a characteristic $\Omega = (\Omega_P, \Omega_\Lambda, \Omega_T)$ is the probability that any fixed plaintext pair satisfying $P' = \Omega_P$ is a right pair when random independent keys are used.

Example of a Characteristic

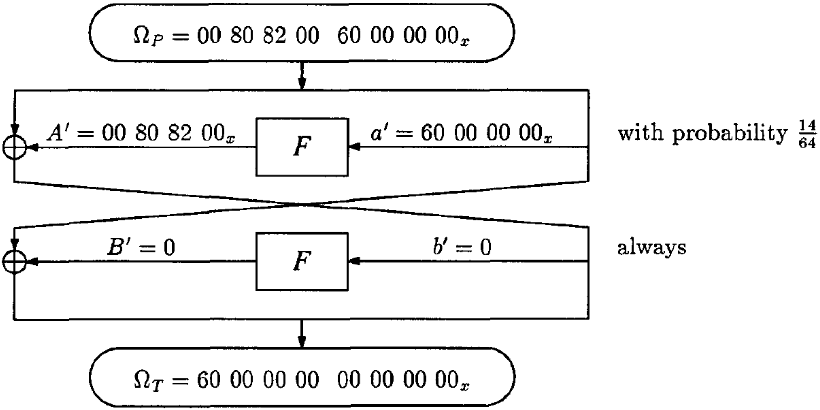

Figure 2: Example of a two-round characteristic with probability $\frac{14}{64}$.

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Definition 6 (Signal-to-Noise Ratio)

The ratio between the number of right pairs and the average count of incorrect subkeys in a counting scheme is called the *signal to noise ratio of the counting scheme* and is denoted by S/N.

Computing the SNR

Consider the variables shown in Table 1.

Variable	Definition
p	Probability of the characteristic
m	Number of created pairs
α	Average count per analyzed pair
β	Fraction of analyzed pairs
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Table 1: Table of variables to compute the SNR.

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Then,

$$S/N = \frac{m \cdot p}{\frac{m \cdot \beta \cdot \alpha}{2^k}} = \frac{2^k \cdot p}{\alpha \cdot \beta} \tag{7}$$

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Definition 7 (Quartet and Octet)

A *quartet* is a structure of four ciphertexts that simultaneously contains two ciphertext pairs of one characteristic and two ciphertext pairs of a second characteristic. An *octet* is a structure of eight ciphertexts that simultaneously contains four ciphertext pairs of each of three characteristics.

3. As an example, $(P, P \oplus \Omega_P^1, P \oplus \Omega_P^2, P \oplus \Omega_P^1 \oplus \Omega_P^2)$ is a quartet.

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4. Quartets save $\frac{1}{2}$ of the data and octets save $\frac{2}{3}$ of the data.

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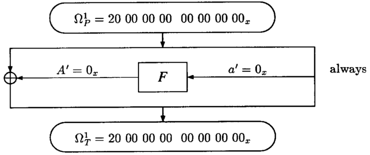

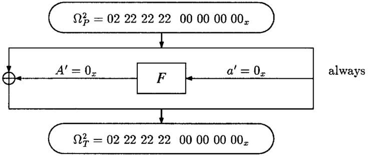

Figure 3: Characteristics used for cryptanalysis of DES reduced to four rounds.

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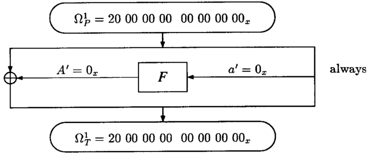
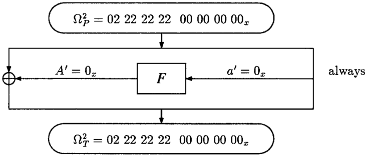
3. Example of a *3R-attack*. There are *three* extra rounds after the characteristic is applied.





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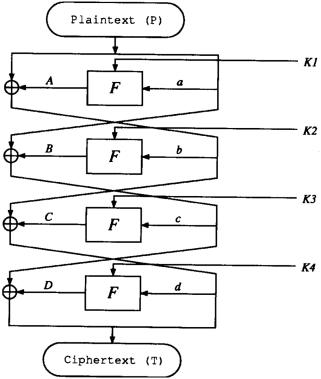

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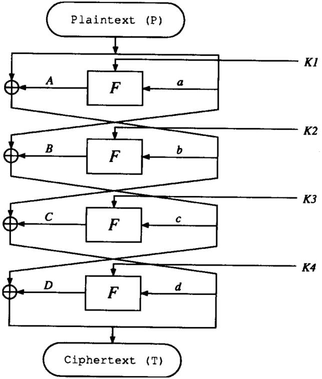

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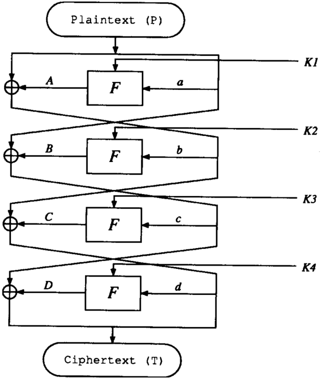

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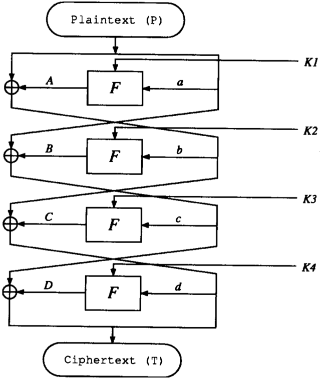

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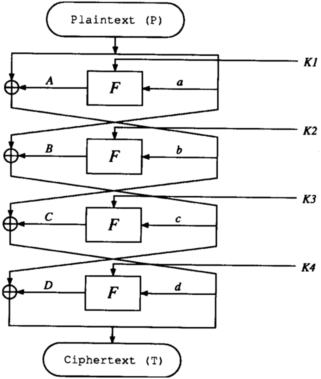

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   - In the second round S2, . . . , S8 receive zero XOR input.
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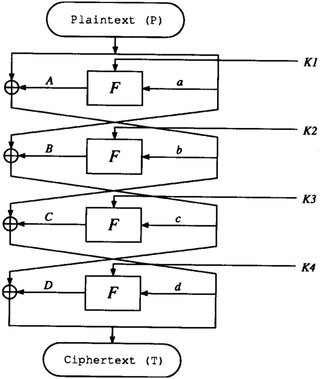
   - We already know $d' = r'$. So, we employ a counting approach to get $K4$.



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DES Reduced to Four Rounds

1. To get Si_{Kd} for $2 \leq i \leq 8$, we verify (9).

$$S(S_{Ed} \oplus S_{Kd}) \oplus S(S^*_{Ed} \oplus S_{Kd}) = S'_{Od} \tag{9}$$

2. Only *one* plaintext pair is needed since characteristic probability is 1.
3. We recover $7 \times 6 = 42$ key bits of $K4$, which correspond to 42 bits of the master key.
4. Exhaustively search the other 14 key bits to get the entire master key.
5. We have used the key schedule to our advantage here? *What if all the keys were independent?*

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3. Since $a' = A' = 0_x$, all keys are equally likely. Other characteristics Ω^3 and Ω^4 are chosen such that
   - $S'_{Ea} \neq 0_x$ for all S boxes for both characteristics.
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   - For every S box, the S'_{Ea} values differ between the characteristics.
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4. 16 chosen plaintexts are needed for this attack.
   - 8 pairs of Ω^1 and Ω^2 each.
   - 4 pairs of Ω^3 and Ω^4 each.

To reduce the data needed, two octets are used.

DES Reduced to Six Rounds

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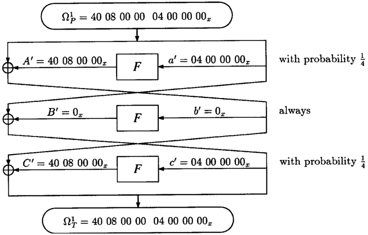

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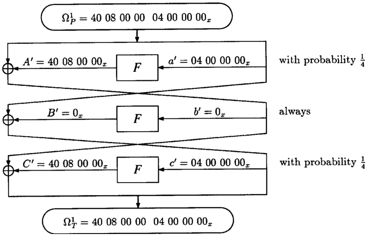
2. We have,

$$e' = c' \oplus D' = F' \oplus l' \implies F' = c' \oplus D' \oplus l' \tag{10}$$



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1. In the fourth round,
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4. Due to higher S/N, fewer plaintext pairs are analyzed. *This is exploited to get a faster counting algorithm*.

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5. Apply this method for both Ω^1 and Ω^2, ensuring that the suggested keys at S2, S5 and S6 match. Otherwise, use more data.

Completing the Cryptanalysis

1. 42 key bits have been found, thus exhaustive search can be performed on the remaining 14 bits.
2. To speed up the search, we can find the remaining 6 key bits of $K6$ using Figure 6. Count using checks on S2, S3 and S8 of the fifth round.
   - Remaining 8 bits can be exhaustively searched.
   - Wrong pairs should be discarded by checking if they satisfy the characteristic and expected value of F'.
   - This will leave us with $\frac{1}{16}$ of the pairs, which boosts S/N greatly.

Into S box number	e bits S_{Ee}	Key bits S_{Ke}
S1	+ + + + + +	**3** + . . + +
S2	+ + **3** + + +	+ **3** + **3** **3** **3**
S3	+ + + + + +	+ + + + + +
S4	+ + + + **3** +	+ + . . + +
S5	**3** + + + + +	+ + + . + +
S6	+ + + + **3** +	+ . + . + +
S7	**3** + + + + +	+ + + . + +
S8	+ + **3** + + +	+ + + + + +

Figure 6: Dependence of $K5$ on bits of $K6$. '3' indicates dependence on $S3_{Kf}$, '.' indicates bits unused in K6 and '+' indicates dependence on known key bits of K6.

Data Requirements

1. The first phase has

$$S/N = \frac{2^{30} \cdot \frac{1}{16}}{4^5} = 2^{16}. \tag{11}$$

Only 7-8 pairs are needed for each characteristic. Since each characteristic has probability $\frac{1}{16}$, we require about 120 pairs of plaintexts.

2. The second phase has

$$S/N = \frac{2^6 \cdot 1}{4} = 16. \tag{12}$$

Though S/N is lesser, we can use the 7-8 right pairs from the first part.

3. We can reduce the data required by using quartets. In total, about 240 ciphertexts are needed.